

# Gamma Glutamyl Transferase

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## Tracking Information

### Publication Number

100-3

### Manual Section Number

190.32

### Manual Section Title

Gamma Glutamyl Transferase

### Version Number

1

### Effective Date of this Version

11/25/2002

### Implementation Date

01/01/2003

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## Description Information

### Benefit Category

Diagnostic Laboratory Tests

**Please Note:** This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

### Item/Service Description

Gamma Glutamyl Transferase (GGT) is an intracellular enzyme that appears in blood following leakage from cells. Renal tubules, liver, and pancreas contain high amounts, although the measurement of GGT in serum is almost always used for assessment of hepatobiliary function. Unlike other enzymes which are found in heart, skeletal muscle, and intestinal mucosa as well as liver, the appearance of an elevated level of GGT in serum is almost always the result of liver disease or injury. It is specifically useful to differentiate elevated alkaline phosphatase levels when the source of the alkaline phosphatase increase (bone, liver, or placenta) is unclear. The combination of high alkaline phosphatase and a normal GGT does not, however, rule out liver disease completely.

As well as being a very specific marker of hepatobiliary function, GGT is also a very sensitive marker for hepatocellular damage. Abnormal concentrations typically appear before

elevations of other liver enzymes or bilirubin are evident. Obstruction of the biliary tract, viral infection (e.g., hepatitis, mononucleosis), metastatic cancer, exposure to hepatotoxins (e.g., organic solvents, drugs, alcohol), and use of drugs that induce microsomal enzymes in the liver (e.g., cimetidine, barbiturates, phenytoin, and carbamazepine) all can cause a moderate to marked increase in GGT serum concentration. In addition, some drugs can cause or exacerbate liver dysfunction (e.g., atorvastatin, troglitazone, and others as noted in FDA Contraindications and Warnings.)

GGT is useful for diagnosis of liver disease or injury, exclusion of hepatobiliary involvement related to other diseases, and patient management during the resolution of existing disease or following injury.

## Indications and Limitations of Coverage

### Indications

1. To provide information about known or suspected hepatobiliary disease, for example:
  - a. Following chronic alcohol or drug ingestion.
  - b. Following exposure to hepatotoxins.
  - c. When using medication known to have a potential for causing liver toxicity (e.g., following the drug manufacturer's recommendations).
  - d. Following infection (e.g., viral hepatitis and other specific infections such as amoebiasis, tuberculosis, psittacosis, and similar infections).
2. To assess liver injury/function following diagnosis of primary or secondary malignant neoplasms.
3. To assess liver injury/function in a wide variety of disorders and diseases known to cause liver involvement (e.g., diabetes mellitus, malnutrition, disorders of iron and mineral metabolism, sarcoidosis, amyloidosis, lupus, and hypertension).
4. To assess liver function related to gastrointestinal disease.
5. To assess liver function related to pancreatic disease.
6. To assess liver function in patients subsequent to liver transplantation.
7. To differentiate between the different sources of elevated alkaline phosphatase activity.

### Limitations

When used to assess liver dysfunction secondary to existing non-hepatobiliary disease with no change in signs, symptoms, or treatment, it is generally not necessary to repeat a GGT determination after a normal result has been obtained unless new indications are present.

If the GGT is the only "liver" enzyme abnormally high, it is generally not necessary to pursue further evaluation for liver disease for this specific indication.

When used to determine if other abnormal enzyme tests reflect liver abnormality rather than other tissue, it generally is not necessary to repeat a GGT more than one time per week.

Because of the extreme sensitivity of GGT as a marker for cytochrome oxidase induction or cell membrane permeability, it is generally not useful in monitoring patients with known liver disease.

Note: Scroll down for links to the quarterly Covered Code Lists (including narrative).

### Cross Reference

Also see the [Medicare Claims Processing Manual](#)<sup>7</sup>, Chapter 120, Clinical Laboratory Services Based on Negotiated Rulemaking.

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## Transmittal Information

### Transmittal Number

## Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R17NCD.pdf> 

## Revision History

07/2004 - Published NCD in the NCD Manual without change to narrative contained in PM AB-02-110. Coding guidance now published in Medicare Lab NCD Manual. Effective and Implementation dates NA. ([TN 17](#) ) (CR 2130)

07/2002 - Implemented NCD. Effective date 11/25/02. Implementation date 1/01/03. ([TN AB-02-110](#) ) (CR 2130)

## Other

### Covered Code Lists (including narrative)

January 2026 (PDF) ([ICD-10](#) )

October 2025 (PDF) ([ICD-10](#) )

January 2025 (PDF) ([ICD-10](#) )

October 2024 (PDF) ([ICD-10](#) )

July 2024 (PDF) ([ICD-10](#) )

April 2024 (PDF) ([ICD-10](#) )

January 2024 (PDF) ([ICD-10](#) )

October 2023 (PDF) ([ICD-10](#) )

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April 2023 (PDF) ([ICD-10](#) )

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July 2022 (PDF) ([ICD-10](#) )

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October 2021 (PDF) ([ICD-10](#) )

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January 2016 ([ICD-10](#))  
October 2015 ([ICD-10](#), [ICD-9](#))  
October 2014 ([ICD-10](#), [ICD-9](#))

## Changes to Lab NCD Edit Software

[January 2026](#)  
[October 2025](#)  
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[October 2014](#)

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## Additional Information

[Other Versions](#)

| Title                      | Version | Effective Between |  |
|----------------------------|---------|-------------------|-----------------------------------------------------------------------------------|
| Gamma Glutamyl Transferase | 1       | 11/25/2002 - N/A  | You are here                                                                      |